

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [19]: bank_df = pd.read_csv("bank_marketing_kaggle_dataset_clean.csv", sep=",")
```

```
In [20]: bank_df.head(10)
```

Out[20]:

	age	job	marital	education	default	balance	housing	loan	contact	day	month	duration	campaign	pdays	previous	poutcome	y
0	30	unemployed	married	primary	no	1787	no	no	cellular	19	oct	79	1	-1	0	unknown	no
1	33	services	married	secondary	no	4789	yes	yes	cellular	11	may	220	1	339	4	failure	no
2	35	management	single	tertiary	no	1350	yes	no	cellular	16	apr	185	1	330	1	failure	no
3	30	management	married	tertiary	no	1476	yes	yes	unknown	3	jun	199	4	-1	0	unknown	no
4	59	blue-collar	married	secondary	no	0	yes	no	unknown	5	may	226	1	-1	0	unknown	no
5	35	management	single	tertiary	no	747	no	no	cellular	23	feb	141	2	176	3	failure	no
6	36	self-employed	married	tertiary	no	307	yes	no	cellular	14	may	341	1	330	2	other	no
7	39	technician	married	secondary	no	147	yes	no	cellular	6	may	151	2	-1	0	unknown	no
8	41	entrepreneur	married	tertiary	no	221	yes	no	unknown	14	may	57	2	-1	0	unknown	no
9	43	services	married	primary	no	-88	yes	yes	cellular	17	apr	313	1	147	2	failure	no

```
In [22]: print(bank_df.columns.tolist())

['age', 'job', 'marital', 'education', 'default', 'balance', 'housing', 'loan', 'contact', 'day', 'month', 'duration', 'campaign', 'pdays', 'previous', 'poutcome', 'y']
```

```
In [21]: bank_df.info()

<class 'pandas.DataFrame'>
RangeIndex: 4521 entries, 0 to 4520
Data columns (total 17 columns):
#   Column      Non-Null Count  Dtype
---  -
0    age         4521 non-null   int64
1    job         4521 non-null   str
2    marital     4521 non-null   str
3    education   4521 non-null   str
4    default     4521 non-null   str
5    balance     4521 non-null   int64
6    housing     4521 non-null   str
7    loan        4521 non-null   str
8    contact     4521 non-null   str
9    day         4521 non-null   int64
10   month       4521 non-null   str
11   duration    4521 non-null   int64
12   campaign    4521 non-null   int64
13   pdays       4521 non-null   int64
14   previous    4521 non-null   int64
15   poutcome    4521 non-null   str
16   y           4521 non-null   str
dtypes: int64(7), str(10)
memory usage: 600.6 KB
```

```
In [38]: bank_df.isnull().sum()
```

Out[38]:

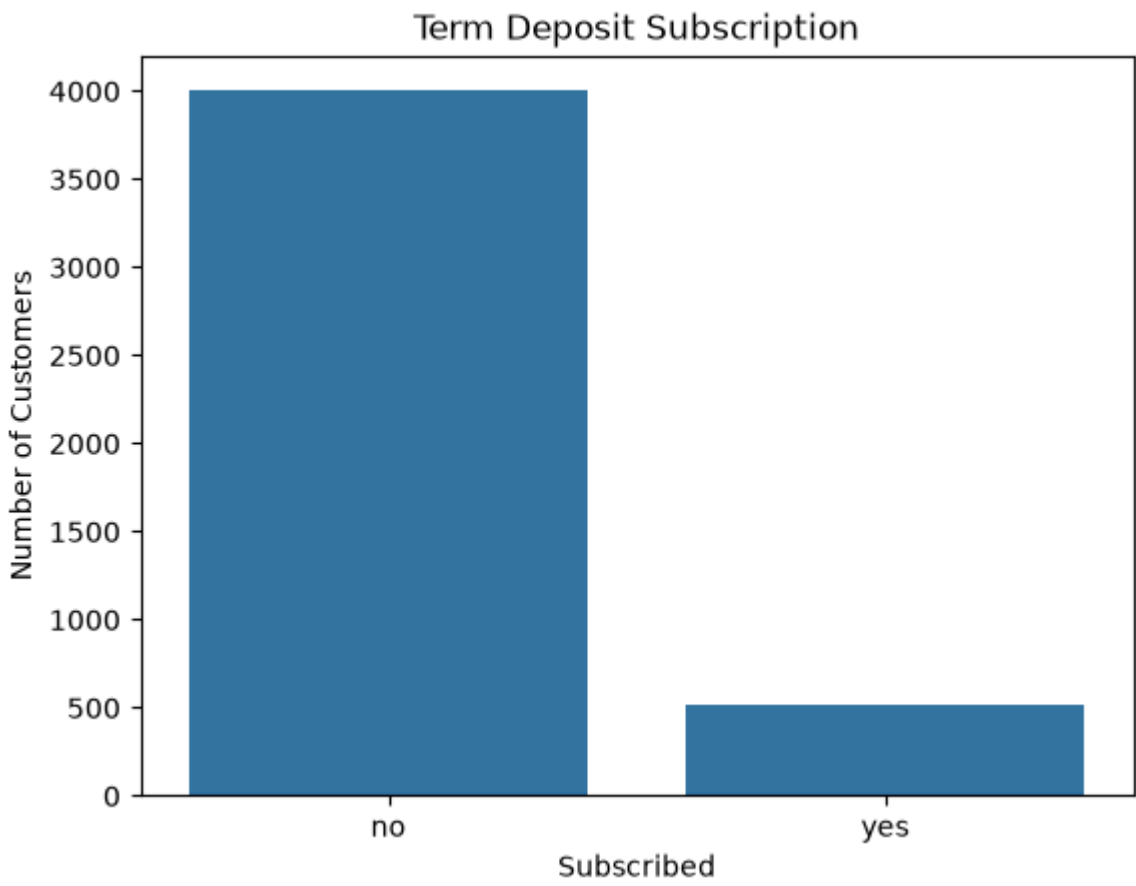
```
age      0
job       0
marital   0
education 0
default   0
balance   0
housing   0
loan      0
contact   0
day       0
month     0
duration  0
campaign  0
pdays    0
previous  0
poutcome  0
y         0
dtype: int64
```

```
In [23]: #Task 2 – Term Deposit Subscription

sns.countplot(x="y", data=bank_df)

plt.title("Term Deposit Subscription")
plt.xlabel("Subscribed")
plt.ylabel("Number of Customers")
```

Out[23]: Text(0, 0.5, 'Number of Customers')



```
In [26]: subscription_percentage = (bank_df["y"].value_counts(normalize=True).mul(100).round(2))

for category, percentage in subscription_percentage.items():
    print(f"{category}: {percentage}%")
```

no: 88.48%
yes: 11.52%

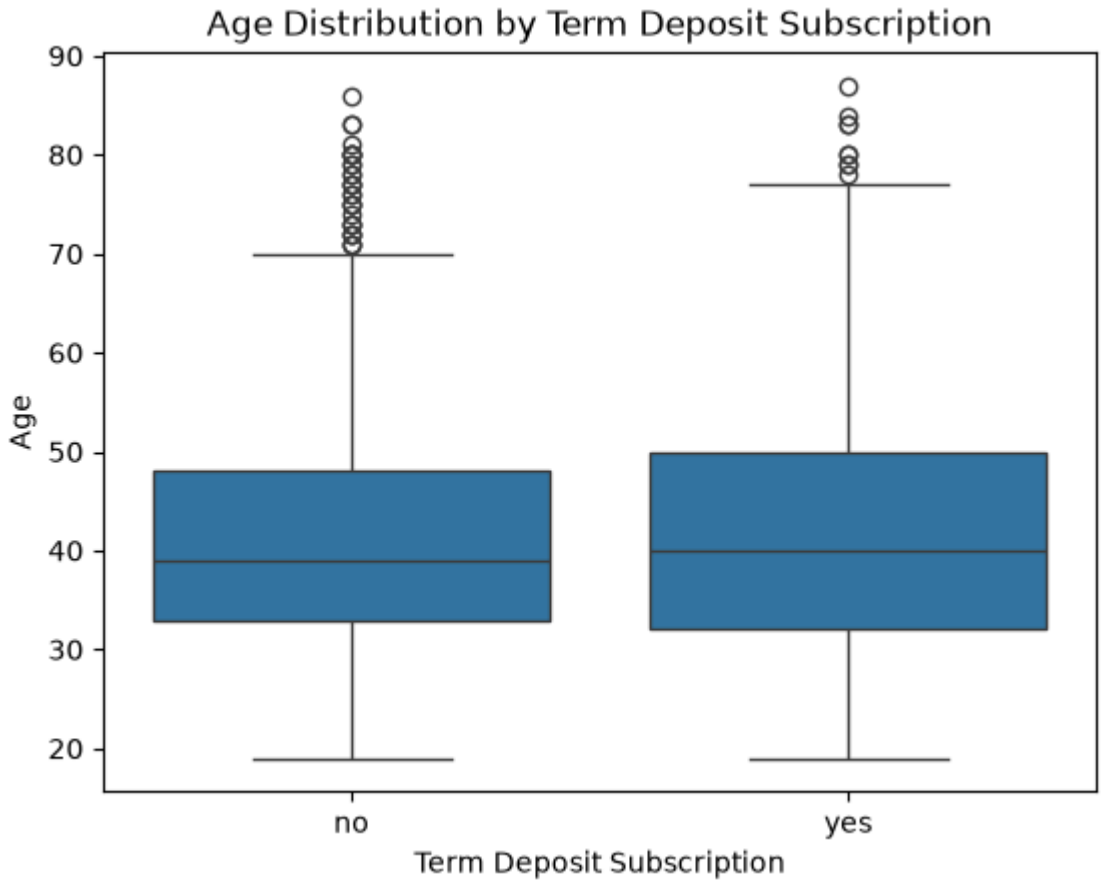
#The majority of customers did not subscribe to the term deposit. Only 11.52% of customers subscribed, while 88.48% did not.

In [27]: #Task 3 – Age Distribution by Subscription

```
sns.boxplot(x="y", y="age", data=bank_df)

plt.title("Age Distribution by Term Deposit Subscription")
plt.xlabel("Term Deposit Subscription")
plt.ylabel("Age")
```

Out[27]: Text(0, 0.5, 'Age')



In [28]: age_summary = bank_df.groupby("y")["age"].agg(Mean_Age="mean", Median_Age="median").round(2)

age_summary

Out[28]:

	Mean_Age	Median_Age
y		
no	41.00	39.0
yes	42.49	40.0

#Conclusion Customers who subscribed to the term deposit had a slightly higher average age than customers who did not subscribe. The mean age was approximately 42.49 years for subscribers compared with 41.00 years for non-subscribers.

In [33]: #Task 4 – Subscription Percentage by Education

```
education_percentage = (bank_df.groupby("education")["y"].apply(lambda x: (x == "yes").mean() * 100).reset_index(name="subscription_percentage"))

education_percentage
```

Out[33]:

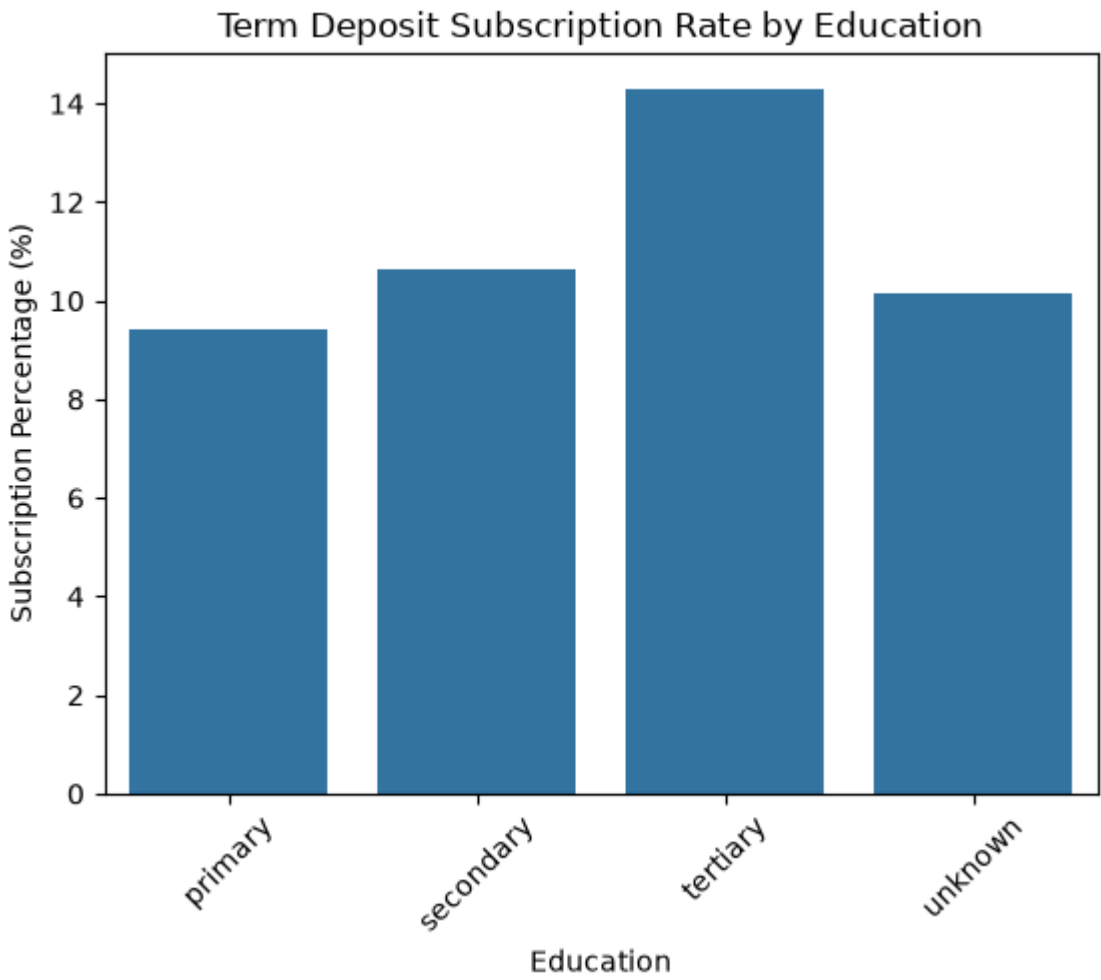
	education	subscription_percentage
0	primary	9.439528
1	secondary	10.624458
2	tertiary	14.296296
3	unknown	10.160428

In [34]: sns.barplot(x="education", y="subscription_percentage", data=education_percentage)

```
plt.title("Term Deposit Subscription Rate by Education")
plt.xlabel("Education")
plt.ylabel("Subscription Percentage (%)")

plt.xticks(rotation=45)
```

Out[34]: ([0, 1, 2, 3],
[Text(0, 0, 'primary'),
Text(1, 0, 'secondary'),
Text(2, 0, 'tertiary'),
Text(3, 0, 'unknown')])



#Conclusion In this dataset, customers with tertiary education had the highest observed term-deposit subscription rate at 14.30%. Customers with primary education had the lowest observed rate at 9.44%.

In [36]: #Task 5 – Additional Visualization

```
job_percentage = (bank_df.groupby("job")["y"].apply(lambda x: (x == "yes").mean() * 100).reset_index(name="subscription_percentage"))

job_percentage
```

Out[36]:

	job	subscription_percentage
0	admin.	12.133891
1	blue-collar	7.293869
2	entrepreneur	8.928571
3	housemaid	12.500000
4	management	13.519092
5	retired	23.478261
6	self-employed	10.928962
7	services	9.112710
8	student	22.619048
9	technician	10.807292
10	unemployed	10.156250
11	unknown	18.421053

In [37]:

```
sns.barplot(x="job", y="subscription_percentage",data=job_percentage)

plt.title("Term Deposit Subscription Rate by Job")
plt.xlabel("Job")
plt.ylabel("Subscription Percentage (%)")

plt.xticks(rotation=45)
```

